

Corruption, Regulation and Firms

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Outline

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1. Introduction

2. Literature

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The impact of corruption on economic activity has long been debated, with two competing perspectives:

- Corruption creates distortions and undermines economic growth (Murphy et al., 1991; Shleifer & Vishny, 1993).
 - Evidence: Mauro (1995); Kaufmann & Wei (1999); Fisman & Svensson (2007)
- Corruption may “grease the wheels” by mitigating frictions in rigid, regulatory bureaucratic systems (Leff, 1964; Huntington, 1968; Lui, 1985).
 - Evidence: Méon & Weill (2010); Vial & Hanoteau (2010); Cai et al. (2011); Bai et al. (2020).

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Motivation (continued)

A key critique of the “grease the wheels” hypothesis is that it treats regulation as completely exogenous, overlooking that corruption may itself give rise to misguided regulation (Bardhan, 1997; Fisman & Golden, 2017; Ding et al., 2020).

- “We have no doubt that quite often delay is deliberately contrived so as to obtain some kind of illicit gratification” (Myrdal, 1968, p. 952, *Asian Drama*).
- “The distortions are not exogenous to the system and are instead often part of the built-in corrupt practices of a patron-client political system” (Bardhan, 1997).

These observations imply that regulatory distortions and corruption are closely linked. Both exogenous and endogenous distortions should be considered when assessing the impact of corruption.

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These observations imply that regulatory distortions and corruption are closely linked. Both exogenous and endogenous distortions should be considered when assessing the impact of corruption.

1. How do corruption and regulation emerge and coexist in a distorted economy?
2. How does corruption affect firm performance, entry, and households' consumption?
3. Are anti-corruption crackdowns always good for firms and households?

We develop a model that jointly characterizes firms' and government's problem:

- Heterogeneous firms choose hiring and bribing decisions to maximize profits.
- Government values local GDP and rent extraction, sets local regulation intensity and firm's connection costs to maximize its objective.
- Corruption, regulation, and stationary distribution of firms are jointly determined in equilibrium.

We discipline the model using Chinese firm-level data for validation and calibration.

We conduct counterfactual analyses to quantify the effects of corruption and anti-corruption crackdowns.

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1. Corruption:

- Theoretical: Leff (1964); Hungtinton (1968); Lui (1985); Murphy et al. (1991); Murphy et al. (1993); Shleifer & Vishny (1993); Grossman & Helpman (1994); Aidt (2003);
- Empirical: Mauro (1995); Kaufmann & Wei (1999); Fisman (2001); Svensson (2003); Faccio (2006); Fisman & Svensson (2007); Méon & Weill (2010); Vial & Hanoteau (2010); Cai et al. (2011); Olken and Pande (2012); Acemoglu et al. (2016); Bai et al. (2020); Fang et al. (2023)
- Anti-corruption crackdown: Chen & Kung (2018); Chen et al. (2020); Giannetti et al. (2021); Griffin et al. (2022); Colonnelli & Prem (2022); Zhang (2023); Colonnelli et al. (2022); Fang & Zhang (2025); Shen et al. (2026)

2. Political connections and large firms: Arayavechkit et al. (2018); Akcigit et al. (2023); Choi et al. (2024); Huneus & Kim (2024)
3. Frictions and Misallocation: Restuccia & Rogerson (2008); Hsieh & Klenow (2009); Bartelsman et al. (2009); Syverson (2011); Hsieh & Klenow (2014)

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1. Standard:

- Steady-state economy.
- Homogeneous goods.
- Competitive goods market.
- DRS technology.
- Inelastic labor supply.
- Endogenous firm entry (ex-ante homogeneous).
- Ex-post heterogeneous firms with idiosyncratic TFP z and sale distortion τ .
- No dynamics in z and τ .

Tailoring:

- Location-specific regulatory intensity ξ generates effective distortion $\xi\tau$.
- Firms can pay f_C to build connections with governments.
- Connected firms pay bribes b shares of output/revenue to offset sale distortion $\xi\tau$.
- Governments value both local GDP and bribe income and are subject to anti-corruption investigations.
- Governments choose ξ and f_C to maximize its objective function.

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Connected firms (type C) with productivity z and effective distortion $\xi\tau$ choose bribing and labor to maximize profits:

$$\pi_C(z, \tau, \xi) = \max_{b,n} (1-b)(1-\xi\tau(1-b))zn^\alpha - wn - f_p,$$

where f_p is the fixed cost of production and w is the wage rate.

For accounting purposes, we define:

- $y_C = (1-b)(1-\xi\tau(1-b))zn^\alpha$ as output sold to households.
- $y_C^g(z, \tau, \xi) = (1-\xi\tau(1-b))zn^\alpha$ as gross output, including the portion consumed by bureaucrats.

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Unconnected Firms

Unconnected firms (NC) with z and $\xi\tau$ cannot bribe governments and have to bear the distortion:

$$\pi_{NC}(z, \tau, \xi) = \max_n (1 - \xi\tau)zn^\alpha - wn - f_P$$

For unconnected firms, $y_{NC} = y_{NC}^g$ because all the output is sold on good market and consumed by households.

The first-order conditions of firms' problems yield testable Propositions:

- **Proposition 1:** A connected firm pays higher bribes when facing greater distortions or tighter regulation: $\frac{\partial b^*}{\partial \tau} > 0$, $\frac{\partial b^*}{\partial \xi} > 0$.
- **Proposition 2:** Firm size (for both connected and unconnected firms) decreases with distortions and regulatory intensity: $\frac{\partial n^*}{\partial \tau} < 0$, $\frac{\partial n^*}{\partial \xi} < 0$.
- **Proposition 3:** Among connected firms, size is negatively correlated with bribes: $\text{corr}(n^*, b^*) < 0$.

We test these propositions using Chinese firm-level data.

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Value Function: Entrants and Type-0 Firms

Entrant firms (type E) are ex ante homogeneous.

Local regulatory intensity ξ is public information.

After paying the entry cost f_E , an entrant draws z and τ from distributions F_z and F_τ , and becomes a type-0 firm with value function $V(z, \tau, \xi)$:

$$V_E(\xi) = -f_E + \mathbb{E}_{z,\tau}[V(z, \tau, \xi)]$$

where

$$V(z, \tau, \xi) = \max\{V_{NC}(z, \tau, \xi), V_C(z, \tau, \xi) - f_C, 0\}.$$

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Value Function: Connected Firms

The value function of a connected firm with (z, τ, ξ) is

$$V_C(z, \tau, \xi) = \pi_C(z, \tau, \xi) + \frac{1}{1+r}(1-\lambda)\left[(1-\eta)V_C(z, \tau, \xi) + \eta V(z, \tau, \xi)\right].$$

- λ is the exogenous firm exit (death) rate.
- $\eta = \psi + p$ is the rate at which a firm loses its government connection, combining:
 1. Exogenous channel: a constant rate ψ capturing routine political turnover.
 2. Endogenous channel: $p(\mathcal{A}, M_C) = 1 - e^{-\mathcal{A} \cdot M_C}$, the probability that the connected bureaucrat is investigated and punished, where $\mathcal{A}$ denotes anti-corruption intensity and $M_C = \int_{z, \tau} \mathbf{1}_C d\mu(z, \tau)$ the mass of connected firms.
 3. Upon losing the connection, the firm reverts to a type-0 firm.

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Value Function: Unconnected Firms

The value function of an unconnected firm is

$$V_{NC}(z, \tau, \xi) = \pi_{NC}(z, \tau, \xi) + \frac{1}{1+r}(1-\lambda)V(z, \tau, \xi).$$

At the end of each period, an unconnected firm reverts to a type-0 firm and determines its type in the next period.

Static Type Choice

Because we abstract from dynamics in z and τ , firms' type choices are effectively static:

- An unconnected firm remains unconnected.

$$V_{NC}(z, \tau, \xi) = \pi_{NC}(z, \tau, \xi) + \frac{1}{1+r}(1-\lambda)V_{NC}(z, \tau, \xi).$$

- A connected firm chooses to reconnect after losing its connection.

$$V_C(z, \tau, \xi) = \pi_C(z, \tau, \xi) + \frac{1}{1+r}(1-\lambda)\left\{ (1-\eta)V_C(z, \tau, \xi) + \eta[V_C(z, \tau, \xi) - f_C] \right\}.$$

- An incumbent firm will not choose to exit the market.

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Operating and Connection Margin

The operating margin is pinned down by unconnected firms $V_{NC}(z, \tau, \xi) \geq 0$, and is equivalent to

$$\pi_{NC}(z, \tau, \xi) = \alpha^{\frac{1}{1-\alpha}} \left(\frac{1}{\alpha} - 1 \right) \left[\frac{(1 - \xi \tau)z}{w^\alpha} \right]^{\frac{1}{1-\alpha}} - f_p \geq 0$$

While the connection margin is disciplined by $V_C(z, \tau, \xi) - f_C \geq V_{NC}(z, \tau, \xi)$, and is equivalent to

$$\begin{aligned} \pi_C(z, \tau, \xi) - \pi_{NC}(z, \tau, \xi) &= \alpha^{\frac{1}{1-\alpha}} \left(\frac{1}{\alpha} - 1 \right) \left(\frac{z}{w^\alpha} \right)^{\frac{1}{1-\alpha}} \left[\left(\frac{1}{4\xi\tau} \right)^{\frac{1}{1-\alpha}} - (1 - \xi\tau)^{\frac{1}{1-\alpha}} \right] \\ &\geq \frac{(1 - \lambda) \cdot \eta(A, M_C) + r + \lambda}{1 + r} f_C(\xi) \end{aligned}$$

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Government's Problem

The government chooses $\{\xi, f_C\}$ to maximize its objective function:

$$G = \max_{\xi, f_C} \lambda B + (1 - \lambda)Y - (1 - e^{-\mathcal{A} \cdot M_C})B^\phi$$

where

- λ : Preference parameter.
- $B = \int_{z, \tau} b(z, \tau, \xi) y_C^g(z, \tau, \xi) d\mu(z, \tau)$: Aggregate bribe income.
- $Y = \int_{z, \tau} y(z, \tau, \xi) d\mu(z, \tau)$: Aggregate household's consumption or GDP.
- $(1 - e^{-\mathcal{A} \cdot M_C})B^\phi$: Expected punishment increasing in $\mathcal{A}$, M_C and B .

1. Corruption has opposing effects on firms:

- Connected firms may benefit from corruption when distortions τ are high:

$$(1 - b)(1 - \xi \tau(1 - b))zn^\alpha \stackrel{?}{\geq} (1 - \tau)zn^\alpha$$

- Depending on the distribution of firms, the “grease-the-wheels” effect may dominate rent-seeking distortions at the aggregate level.

2. The effect of anti-corruption intensity $\mathcal{A}$ on bureaucrats is also ambiguous:

- For low $\mathcal{A}$, governments may reduce connected firms by increasing f_C without adjusting ξ . The “grease-the-wheels” channel is weakened, while rent-seeking distortions persist.
- For sufficiently high $\mathcal{A}$, governments reduce ξ as the cost of corruption becomes too large; however, connected firms can no longer offset high distortions τ .

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1. Aggregate output:

$$Y^g = \int_{z,\tau} y^g(z, \tau, \xi) d\mu(z, \tau)$$

where $\mu(z, \tau)$ denote the distribution of operating firms.

2. Profit transfer:

$$\Pi = \int_{z,\tau} \pi(z, \tau, \xi) d\mu(z, \tau)$$

3. Accounting:

$$Y^g = B + Y, Y = WL + \Pi$$

4. Labor market clearing:

$$L = \int_{z,\tau} n d\mu(z, \tau)$$

Definition of Equilibrium

A steady-state competitive equilibrium consists of $(w, \xi, f_C, E, M_C, \mu)$, value functions, policy functions such that:

- Firms' policy and value functions solve their optimization problems.
- The government chooses (ξ, f_C) to maximize its objective function.
- Labor market clears: $L = \int n(z, \tau, \xi) d\mu(z, \tau)$.
- Free entry holds: $V_E(\xi) = 0$.
- Aggregation and accounting hold for Y , Y^g , and B
- The distribution $\mu(z, \tau)$ is invariant given entry, exit, and firms' type choices.

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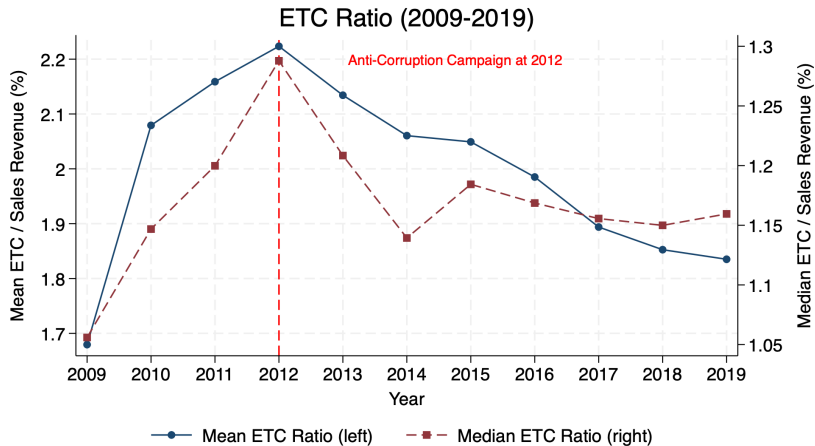
5. Appendix

- Weak formal institution in China generates distortions for private firms (Acemoglu & Robinson, 2012; Bai et al., 2020).
- China's economically decentralized regime creates variation in location-specific policies and regulations (Xu, 2011).
- Corruption is widespread and often viewed as “greasing the wheels” for private firms (Cai et al., 2011; Bai et al., 2020), though its relationship with regulation remains ambiguous (Ding et al., 2020).
- The literature documents mixed effects of anti-corruption crackdowns in China:
 - The 2004 Heilongjiang provincial crackdown reduced firm productivity and entry (Chen et al., 2020).
 - Xi Jinping's 2012 anti-corruption campaign is considered unprecedented and improved firm performance and entry (Ding et al., 2020; Giannetti et al., 2021).

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Firm-level data:

- Source: *China Stock Market and Accounting Research* (CSMAR)
- Period: 2009–2019
- Firm-year observations: 20,400
- Corruption measure: entertainment and travel cost (ETC) based on Cai et al. (2011)
- Other variables: total assets, employees, ROA, leverage, capital intensity, firm age, and an SOE indicator



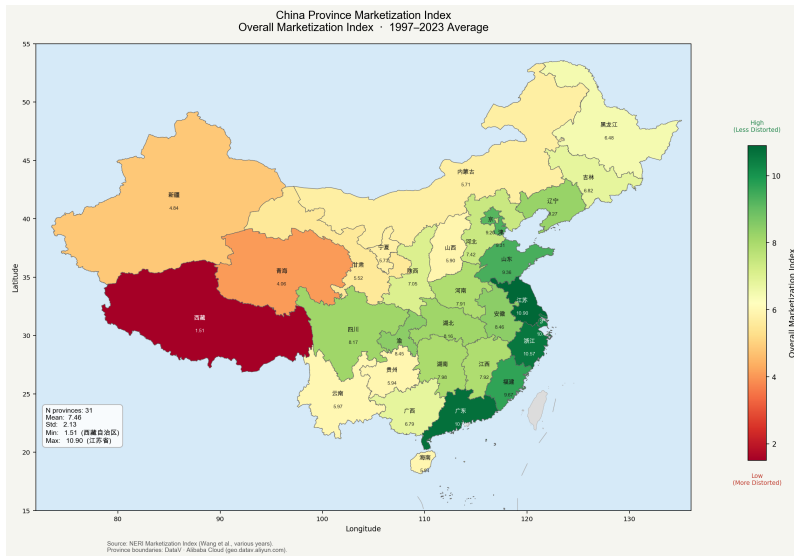
Note: Data is winsorized at 1% and 99% levels.

FIGURE : ETC trend before and after the anti-corruption campaign

Provincial distortion data:

- Source: *NERI China Marketization Index*
- Period: 1997-2023
- The overall marketization index consists of five subindices:
 1. Government-market relation index
 2. Development of non-state-owned sector
 3. Development of product market
 4. Development of factor market
 5. Legal Protection

Heterogeneity in Provincial Marketization



1. **Provincial distortion** ξ : proxied by the government–market relationship index, which captures government intervention, firm tax burdens, and the extent of market-based resource allocation.
 - We invert and standardize the index so that higher values indicate greater regulatory distortion:

$$\tilde{\xi}_{pt} = \frac{-\widehat{\text{GovtMkt}}_{pt} - \overline{(-\widehat{\text{GovtMkt}})}}{\sigma}$$

2. **Industry distortion** τ_{jt} : measured as the industry-year average tax burden,

$$\tau_{jt} = \frac{1}{N_{jt}} \sum_{i \in j} \tau_{it},$$

where τ_{it} is firm i 's tax-to-revenue ratio. As an industry-level aggregate, τ_{jt} is plausibly exogenous to any single firm's bribery behavior.

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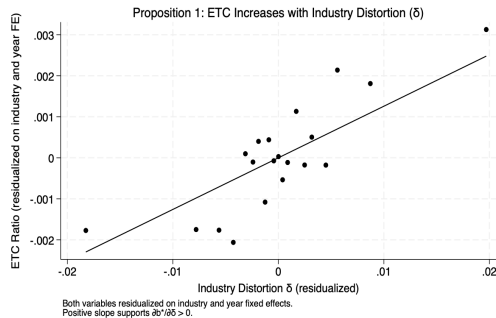
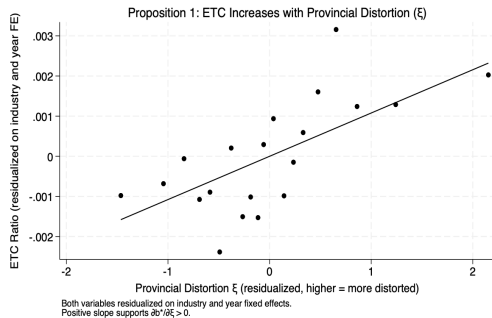
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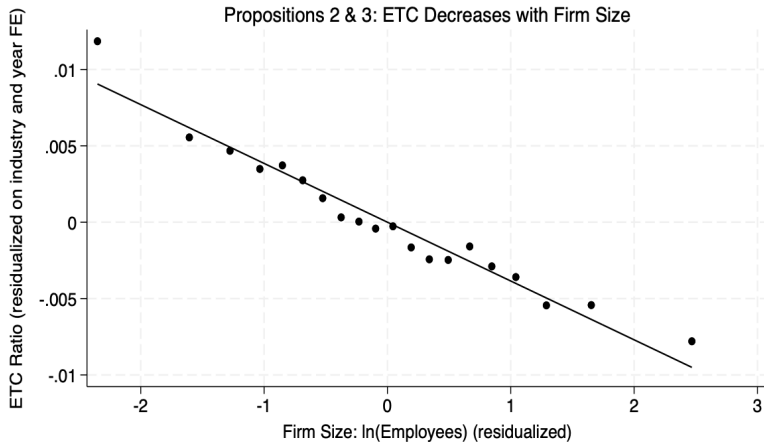
Test of Proposition 1



Left: ETC vs. Provincial Distortion ($\tilde{\xi}$) **Right:** ETC vs. Industry Distortion (τ)

Residualized on industry and year FE. Positive slopes support Proposition 1.

Test of Proposition 3



Both variables residualized on industry and year fixed effects.
Negative slope supports $\partial n^* / \partial (\xi \delta) < 0$ (Prop 2)
and $\text{corr}(n^*, b^*) < 0$ (Prop 3).

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4. Empirical Tests

5. Appendix

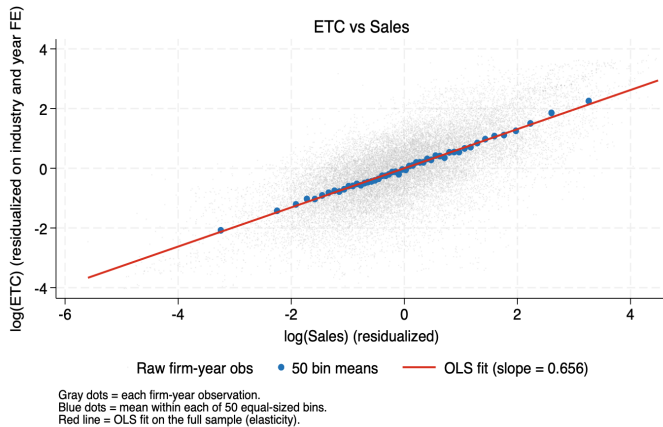


FIGURE : ETC and Sales. Source: CSMAR

Examples of Sale Distortions τ

Sales distortions τ encompass a range of policy- and institution-driven frictions, including taxes, regulatory burdens, compliance costs (e.g., paperwork), production quotas, and delays or inefficiencies in bureaucratic processes such as obtaining permits. Existing studies include:

- **Cai et al. (2011):** Higher tax burden and weaker government support are associated with lower TFPR among Chinese firms.
- **Kalmenovitz (2023):** Firms facing higher regulatory intensity allocate a larger share of revenue to SGA and COGS.
- **Crane and Koch (2025):** Regulatory intensity reduces firm valuations, except for politically connected firms.

Examples of Location-Specific ξ

Although τ (taxes, regulations, quotas, etc.) is pervasive, there is substantial heterogeneity in how governments implement these policies across locations ξ .

- The World Bank's *Doing Business* reports document large cross-country differences in regulatory burdens (e.g., starting a business, dealing with construction permits, paying taxes).
 - For example, the time required to obtain a construction permit ranges from 26 to 373 days across countries in 2020.
- Using the marketisation index constructed by Wang et al. (2023) for China (1997–2023), we observe significant cross-province heterogeneity in institutional quality, including government–market relations and the development of factor and product markets.

Examples of Bribery Behavior

We provide examples showing how firms use bribery or lobbying b to adjust their sales distortions:

- **Cai et al. (2011):** Firms facing higher tax burdens and weaker government support are more likely to engage in bribery; conditional on these factors, bribery is associated with higher TFPR.
- **Amin & Soh (2021):** Using cross-country firm-level data, higher regulatory burden increases firms' bribery payments.
- **Kalmenovitz (2023):** U.S. firms facing higher regulatory intensity engage more in lobbying activities.
- **Huneus & Kim (2024):** Explicitly model firms' lobbying behavior as a tool to adjust sale distortions.